

# Interventions & Events

AE, CM and EX — the general observation classes

DM was one row per subject. From here on, subjects have MANY records — which means sequence numbers, study days, and a lot more timing.

# By the end of this module you can...

1

**Tell Interventions from Events** and know which domains fall into each class.

2

**Read the general-observation structure** topic, qualifiers and timing — the pattern every non-Special-Purpose domain follows.

3

**Explain AETERM vs AEDECOD** and why a submission keeps both the verbatim and the coded term.

4

**Distinguish severity from seriousness** the single most confused pair in the whole standard.

5

**Derive --SEQ and --DY** the two variables DM never needed.

# Done TO the subject, or happened TO the subject?

## INTERVENTIONS

Something given to, or done to, the subject — deliberately.

**EX** Exposure — the study drug itself

**CM** Concomitant medications — everything else they took

**PR** Procedures

Topic variable: `--TRT` (what was given)

`EXTRT` · `CMTRT`

## EVENTS

Something that happened to the subject — not planned.

**AE** Adverse events — the safety backbone

**MH** Medical history — conditions before the study

**DS** Disposition — milestones and how they left

Topic variable: `--TERM` (what happened)

`AETERM` · `MHTERM` · `DSTERM`

*The test: was it deliberate (Intervention) or did it just happen (Event)? Aspirin taken for a headache is an Intervention; the headache is an Event.*

## THE PATTERN

# Every general-observation domain has the same skeleton

1

### IDENTIFIERS

Who and which record

STUDYID · DOMAIN · USUBJID · --SEQ

2

### TOPIC

What the record is about

AETERM · CMTRT · EXTRT

3

### QUALIFIERS

Everything describing the topic

AESEV · AESER · AEREL · AEOUT · AEDECOD

4

### TIMING

When it happened

AESTDTC · AEENDTC · AESTDY · AEENDY

**This is what DM did NOT have.** DM is Special Purpose — no topic variable, no --SEQ. From here on every domain follows the skeleton above, which is why learning it once pays off six times.

# Anatomy of an AE record

| USUBJID       | AESEQ | AETERM       | AEDECOD  | AESEV    | AESER | AESTDTC    | AEENDTC    | AESTDY |
|---------------|-------|--------------|----------|----------|-------|------------|------------|--------|
| ABC-01-01-001 | 1     | bad headache | Headache | MODERATE | N     | 2024-03-15 | 2024-03-16 | 15     |

Identifiers

USUBJID + AESEQ uniquely identify this row

Topic

AETERM — what the investigator reported, verbatim

Qualifiers

AEDECOD, AESEV, AESER, AEREL, AEOUT describe it

Timing

AESTDTC / AEENDTC in ISO, AESTDY / AEENDY as study days

One row per event, per subject. A subject with three adverse events gets three rows — AESEQ 1, 2, 3.

TWO TERMS, BOTH REQUIRED

# AETERM and AEDECOD

## AETERM

VERBATIM — exactly what the investigator wrote.

"bad headache"

Never cleaned, never corrected, never abbreviated. It is the legal record of what was reported.



## AEDECOD

CODED — the dictionary's preferred term.

"Headache"

Assigned by trained coders using MedDRA. Lets you count how many subjects had headache across the whole study.

**Why keep both?** The verbatim preserves what actually happened; the coded term makes it countable. Three investigators writing "bad headache", "headache" and "cephalalgia" all code to one preferred term — and only then can you say how many subjects had it.

⚠ **Coding is not a lookup table.** MedDRA is a licensed dictionary applied by trained coders — you cannot derive AEDECOD with a CASE statement. In this bootcamp the coded values are ILLUSTRATIVE stand-ins so the exercise runs.

# Severity is NOT seriousness

## AESEV — Severity

How BAD did it feel?

MILD · MODERATE · SEVERE

A clinical judgement of intensity.

## AESER — Seriousness

Did it meet a REGULATORY threshold?

Y / N

Death, life-threatening, hospitalisation, disability, congenital anomaly, or medically important.

A mild rash that required hospitalisation

MILD

Y

Serious but not severe

Severe pain, resolved at home

SEVERE

N

Severe but not serious

Worsening hypertension, subject withdrawn

SEVERE

Y

Both — our ABC-01 case

They are independent. Never derive one from the other — a serious event is not automatically severe, and vice versa.

# --SEQ: numbering a subject's records

DM never needed this. Now that a subject can have many rows in one domain, every row needs a unique number within that subject.

| USUBJID       | AESEQ | AETERM                 | AESTDTC    |
|---------------|-------|------------------------|------------|
| ABC-01-01-001 | 1     | bad headache           | 2024-03-15 |
| ABC-01-01-001 | 2     | Nausea                 | 2024-03-20 |
| ABC-01-01-004 | 1     | fatigue                | 2024-03-18 |
| ABC-01-01-004 | 2     | worsening hypertension | 2024-03-22 |

### The rule

Sequential 1, 2, 3... WITHIN each subject. It restarts at 1 for the next subject — it is not a row number for the whole dataset.

### The test

USUBJID + AESEQ must be UNIQUE. That pair is what identifies any single observation in the study.

Sort deterministically before you number — otherwise the same program gives different --SEQ values on different runs.



# --DY: study day, and the missing Day 0

## The rule

```
if  --DTC >= RFSTDTC :  --DY = (--DTC - RFSTDTC) + 1      <- on/after first dose
if  --DTC < RFSTDTC :  --DY = (--DTC - RFSTDTC)           <- before, stays negative
```

First dose

2024-03-01

**Day 1**

Headache starts

2024-03-15

**Day 15**

Screening visit

2024-02-20

**Day -10**

**There is no Day 0.** The day before first dose is Day -1. If your code ever produces a 0, the +1 is missing on one branch.

# Ongoing events

Two of the nine ABC-01 adverse events had not resolved when the data was collected. Their end date is blank — and it must stay blank.

|                |                            |                                                                   |
|----------------|----------------------------|-------------------------------------------------------------------|
| <b>AEENDTC</b> | Leave NULL                 | Do not substitute today's date, the last visit, or the study end. |
| <b>AEENDY</b>  | Leave NULL                 | No end date means no end study day. Deriving one invents data.    |
| <b>AEOUT</b>   | NOT RECOVERED/NOT RESOLVED | The outcome carries the information that it is ongoing.           |

**Why this matters.** Inventing an end date would make an unresolved event look resolved. In a safety review that is the difference between an ongoing problem and a closed one — which is exactly the kind of thing a regulator is looking for.

PUTTING IT TOGETHER

raw AE → SDTM AE

**AETERM**

verbatim, unchanged

bad headache → bad headache

**AESTDTC**

MIXED formats → ISO 8601

15/03/2024 → 2024-03-15

**AESER**

normalise → N / Y

No, N → N

**AESQ**

derive per subject

1, 2, 3 ...

**AEDECOD**

dictionary coded (MedDRA)

bad headache → Headache

**AESEV**

trim + upper-case → CT

moderate → MODERATE

**AEOUT**

DECODE the number → CT

1 → RECOVERED/RESOLVED

**AESTDY**

derive from RFSTDTC

2024-03-15 → 15

The AE date conversion is the hardest in the study: ONE column contains two different formats. You must detect the format per value.

# Common AE mistakes



## **Cleaning up AETERM**

The verbatim term is a legal record. Fix nothing — not spelling, not capitalisation.



## **Deriving AEDECOD in code**

MedDRA coding is done by trained coders with a licensed dictionary.



## **Confusing severity and seriousness**

They are independent. A mild event can be serious.



## **Assuming one date format**

AESTDT holds BOTH DD/MM/YYYY and DD-Mon-YYYY. Detect per value.



## **Filling in a blank end date**

Ongoing means ongoing. Null AEENDTC and null AEENDY.



## **Numbering --SEQ across subjects**

AESEQ restarts at 1 for each subject; it is not a row number.

WHAT'S NEXT

# Build AE, then the Interventions

1

**Notebook 05 • AE** Mixed dates, CT, the AEOUT code decode, AESEQ and AESTDY — in SAS.

2

**Day 6 • CM and EX** Same skeleton, Interventions instead of Events. Much of your AE code transfers.

3

**Then Findings** VS and LB — where the structure changes shape again.

*AE is the hardest domain you will build. Everything after it reuses what you learn here.*